MSG No. 219.87 $F\bar{4}'3c'$ [Type III, cubic]

Table 1: Wyckoff site: `8a`, site symmetry: `23.`

No.	position	mapping
1	$[0, 0, 0]$	`[1,2,3,4,5,6,7,8,9,10,11,12]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[13,14,15,16,17,18,19,20,21,22,23,24]`
3	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[25,26,27,28,29,30,31,32,33,34,35,36]`
4	$\left[\frac{1}{2}, 0, 0\right]$	`[37,38,39,40,41,42,43,44,45,46,47,48]`
5	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[49,50,51,52,53,54,55,56,57,58,59,60]`
6	$\left[0, \frac{1}{2}, 0\right]$	`[61,62,63,64,65,66,67,68,69,70,71,72]`
7	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[73,74,75,76,77,78,79,80,81,82,83,84]`
8	$\left[0, 0, \frac{1}{2}\right]$	`[85,86,87,88,89,90,91,92,93,94,95,96]`

Table 2: Wyckoff site: `8b`, site symmetry: `23.`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[1,5,6,26,34,35,51,55,60,76,80,81]`
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[2,10,11,25,29,30,52,56,57,75,79,84]`
3	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[3,7,12,28,32,33,49,53,54,74,82,83]`
4	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[4,8,9,27,31,36,50,58,59,73,77,78]`
5	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[13,18,23,38,39,43,68,70,72,88,89,93]`
6	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[14,15,19,37,42,47,64,65,69,92,94,96]`
7	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[16,17,21,44,46,48,62,63,67,85,90,95]`
8	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[20,22,24,40,41,45,61,66,71,86,87,91]`

Table 3: Wyckoff site: `24c`, site symmetry: `-4'..`

No.	position	mapping
1	$\left[0, \frac{1}{4}, \frac{1}{4}\right]$	`[1,26,62,85]`
2	$\left[0, \frac{3}{4}, \frac{3}{4}\right]$	`[2,25,61,86]`
3	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	`[3,28,69,94]`
4	$\left[0, \frac{3}{4}, \frac{1}{4}\right]$	`[4,27,70,93]`
5	$\left[\frac{1}{4}, 0, \frac{1}{4}\right]$	`[5,43,60,89]`
6	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	`[6,47,64,81]`
7	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[7,48,63,83]`
8	$\left[\frac{3}{4}, 0, \frac{1}{4}\right]$	`[8,44,58,90]`
9	$\left[\frac{3}{4}, \frac{3}{4}, 0\right]$	`[9,40,71,78]`
10	$\left[\frac{1}{4}, 0, \frac{3}{4}\right]$	`[10,42,56,92]`
11	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[11,39,72,79]`
12	$\left[\frac{3}{4}, 0, \frac{3}{4}\right]$	`[12,41,53,91]`
13	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	`[13,38,50,73]`
14	$\left[\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	`[14,37,49,74]`
15	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[15,35,55,96]`
16	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[16,33,54,95]`
17	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{4}\right]$	`[17,36,67,77]`
18	$\left[\frac{1}{4}, \frac{1}{2}, \frac{1}{4}\right]$	`[18,34,68,80]`

continued ...

Table 3

No.	position	mapping
19	$\left[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}\right]$	`[19,29,65,84]`
20	$\left[\frac{3}{4}, \frac{1}{2}, \frac{3}{4}\right]$	`[20,32,66,82]`
21	$\left[\frac{1}{2}, \frac{1}{4}, \frac{1}{4}\right]$	`[21,46,51,76]`
22	$\left[\frac{1}{2}, \frac{3}{4}, \frac{3}{4}\right]$	`[22,45,52,75]`
23	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[23,30,57,88]`
24	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[24,31,59,87]`

Table 4: Wyckoff site: `24d`, site symmetry: `-4'..`

No.	position	mapping
1	$\left[\frac{1}{4}, 0, 0\right]$	`[1,2,37,38]`
2	$\left[\frac{3}{4}, 0, 0\right]$	`[3,4,45,46]`
3	$\left[0, \frac{1}{4}, 0\right]$	`[5,12,65,67]`
4	$\left[0, 0, \frac{1}{4}\right]$	`[6,9,88,95]`
5	$\left[0, 0, \frac{3}{4}\right]$	`[7,11,87,96]`
6	$\left[0, \frac{3}{4}, 0\right]$	`[8,10,66,68]`
7	$\left[\frac{1}{4}, \frac{1}{2}, \frac{1}{2}\right]$	`[13,14,25,26]`
8	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[15,24,79,83]`
9	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	`[16,23,78,81]`
10	$\left[\frac{1}{2}, \frac{1}{4}, \frac{1}{2}\right]$	`[17,19,53,60]`
11	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{2}\right]$	`[18,20,56,58]`
12	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{2}\right]$	`[21,22,27,28]`
13	$\left[0, \frac{3}{4}, \frac{1}{2}\right]$	`[29,36,89,91]`
14	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[30,33,64,71]`
15	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[31,35,63,72]`
16	$\left[0, \frac{1}{4}, \frac{1}{2}\right]$	`[32,34,90,92]`
17	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[39,48,55,59]`
18	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[40,47,54,57]`
19	$\left[\frac{1}{2}, \frac{3}{4}, 0\right]$	`[41,43,77,84]`
20	$\left[\frac{1}{2}, \frac{1}{4}, 0\right]$	`[42,44,80,82]`
21	$\left[\frac{3}{4}, 0, \frac{1}{2}\right]$	`[49,50,85,86]`
22	$\left[\frac{1}{4}, 0, \frac{1}{2}\right]$	`[51,52,93,94]`
23	$\left[\frac{3}{4}, \frac{1}{2}, 0\right]$	`[61,62,73,74]`
24	$\left[\frac{1}{4}, \frac{1}{2}, 0\right]$	`[69,70,75,76]`

Table 5: Wyckoff site: `32e`, site symmetry: `.3.`

No.	position	mapping
1	$[x, x, x]$	`[1,5,6]`
2	$[x, -x, -x]$	`[2,10,11]`
3	$[-x, x, -x]$	`[3,7,12]`
4	$[-x, -x, x]$	`[4,8,9]`
5	$\left[\frac{1}{2} - x, x + \frac{1}{2}, \frac{1}{2} - x\right]$	`[13,18,23]`
6	$\left[\frac{1}{2} - x, \frac{1}{2} - x, x + \frac{1}{2}\right]$	`[14,15,19]`

continued ...

Table 5

No.	position	mapping
7	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}-x\right]$	[16,17,21]
8	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ x+\frac{1}{2}\right]$	[20,22,24]
9	$\left[x,\ x+\frac{1}{2},\ x+\frac{1}{2}\right]$	[25,29,30]
10	$\left[x,\ \frac{1}{2}-x,\ \frac{1}{2}-x\right]$	[26,34,35]
11	$\left[-x,\ x+\frac{1}{2},\ \frac{1}{2}-x\right]$	[27,31,36]
12	$\left[-x,\ \frac{1}{2}-x,\ x+\frac{1}{2}\right]$	[28,32,33]
13	$\left[\frac{1}{2}-x,\ x,\ -x\right]$	[37,42,47]
14	$\left[\frac{1}{2}-x,\ -x,\ x\right]$	[38,39,43]
15	$\left[x+\frac{1}{2},\ -x,\ -x\right]$	[40,41,45]
16	$\left[x+\frac{1}{2},\ x,\ x\right]$	[44,46,48]
17	$\left[x+\frac{1}{2},\ x,\ x+\frac{1}{2}\right]$	[49,53,54]
18	$\left[x+\frac{1}{2},\ -x,\ \frac{1}{2}-x\right]$	[50,58,59]
19	$\left[\frac{1}{2}-x,\ x,\ \frac{1}{2}-x\right]$	[51,55,60]
20	$\left[\frac{1}{2}-x,\ -x,\ x+\frac{1}{2}\right]$	[52,56,57]
21	$\left[-x,\ x+\frac{1}{2},\ -x\right]$	[61,66,71]
22	$\left[-x,\ \frac{1}{2}-x,\ x\right]$	[62,63,67]
23	$\left[x,\ \frac{1}{2}-x,\ -x\right]$	[64,65,69]
24	$\left[x,\ x+\frac{1}{2},\ x\right]$	[68,70,72]
25	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ x\right]$	[73,77,78]
26	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ -x\right]$	[74,82,83]
27	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ -x\right]$	[75,79,84]
28	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ x\right]$	[76,80,81]
29	$\left[-x,\ x,\ \frac{1}{2}-x\right]$	[85,90,95]
30	$\left[-x,\ -x,\ x+\frac{1}{2}\right]$	[86,87,91]
31	$\left[x,\ -x,\ \frac{1}{2}-x\right]$	[88,89,93]
32	$\left[x,\ x,\ x+\frac{1}{2}\right]$	[92,94,96]

Table 6: Wyckoff site: 48f, site symmetry: 2..

No.	position	mapping
1	$[x,\ 0,\ 0]$	[1,2]
2	$[-x,\ 0,\ 0]$	[3,4]
3	$[0,\ x,\ 0]$	[5,12]
4	$[0,\ 0,\ x]$	[6,9]
5	$[0,\ 0,\ -x]$	[7,11]
6	$[0,\ -x,\ 0]$	[8,10]
7	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	[13,14]
8	$\left[\frac{1}{2},\ \frac{1}{2},\ x+\frac{1}{2}\right]$	[15,24]
9	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-x\right]$	[16,23]
10	$\left[\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	[17,19]
11	$\left[\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}\right]$	[18,20]
12	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	[21,22]
13	$\left[x,\ \frac{1}{2},\ \frac{1}{2}\right]$	[25,26]
14	$\left[-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	[27,28]
15	$\left[0,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	[29,36]

continued ...

Table 6

No.	position	mapping
16	$\left[0,\ \frac{1}{2},\ x+\frac{1}{2}\right]$	[30,33]
17	$\left[0,\ \frac{1}{2},\ \frac{1}{2}-x\right]$	[31,35]
18	$\left[0,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	[32,34]
19	$\left[\frac{1}{2}-x,\ 0,\ 0\right]$	[37,38]
20	$\left[\frac{1}{2},\ 0,\ x\right]$	[39,48]
21	$\left[\frac{1}{2},\ 0,\ -x\right]$	[40,47]
22	$\left[\frac{1}{2},\ -x,\ 0\right]$	[41,43]
23	$\left[\frac{1}{2},\ x,\ 0\right]$	[42,44]
24	$\left[x+\frac{1}{2},\ 0,\ 0\right]$	[45,46]
25	$\left[x+\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	[49,50]
26	$\left[\frac{1}{2}-x,\ 0,\ \frac{1}{2}\right]$	[51,52]
27	$\left[\frac{1}{2},\ x,\ \frac{1}{2}\right]$	[53,60]
28	$\left[\frac{1}{2},\ 0,\ x+\frac{1}{2}\right]$	[54,57]
29	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-x\right]$	[55,59]
30	$\left[\frac{1}{2},\ -x,\ \frac{1}{2}\right]$	[56,58]
31	$\left[-x,\ \frac{1}{2},\ 0\right]$	[61,62]
32	$\left[0,\ \frac{1}{2},\ x\right]$	[63,72]
33	$\left[0,\ \frac{1}{2},\ -x\right]$	[64,71]
34	$\left[0,\ \frac{1}{2}-x,\ 0\right]$	[65,67]
35	$\left[0,\ x+\frac{1}{2},\ 0\right]$	[66,68]
36	$\left[x,\ \frac{1}{2},\ 0\right]$	[69,70]
37	$\left[x+\frac{1}{2},\ \frac{1}{2},\ 0\right]$	[73,74]
38	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ 0\right]$	[75,76]
39	$\left[\frac{1}{2},\ x+\frac{1}{2},\ 0\right]$	[77,84]
40	$\left[\frac{1}{2},\ \frac{1}{2},\ x\right]$	[78,81]
41	$\left[\frac{1}{2},\ \frac{1}{2},\ -x\right]$	[79,83]
42	$\left[\frac{1}{2},\ \frac{1}{2}-x,\ 0\right]$	[80,82]
43	$\left[-x,\ 0,\ \frac{1}{2}\right]$	[85,86]
44	$\left[0,\ 0,\ x+\frac{1}{2}\right]$	[87,96]
45	$\left[0,\ 0,\ \frac{1}{2}-x\right]$	[88,95]
46	$\left[0,\ -x,\ \frac{1}{2}\right]$	[89,91]
47	$\left[0,\ x,\ \frac{1}{2}\right]$	[90,92]
48	$\left[x,\ 0,\ \frac{1}{2}\right]$	[93,94]

Table 7: Wyckoff site: 48g, site symmetry: 2..

No.	position	mapping
1	$\left[x,\ \frac{1}{4},\ \frac{1}{4}\right]$	[1,26]
2	$\left[x,\ \frac{3}{4},\ \frac{3}{4}\right]$	[2,25]
3	$\left[-x,\ \frac{1}{4},\ \frac{3}{4}\right]$	[3,28]
4	$\left[-x,\ \frac{3}{4},\ \frac{1}{4}\right]$	[4,27]
5	$\left[\frac{1}{4},\ x,\ \frac{1}{4}\right]$	[5,60]
6	$\left[\frac{1}{4},\ \frac{1}{4},\ x\right]$	[6,81]
7	$\left[\frac{3}{4},\ \frac{1}{4},\ -x\right]$	[7,83]
8	$\left[\frac{3}{4},\ -x,\ \frac{1}{4}\right]$	[8,58]

continued ...

Table 7

No.	position	mapping
9	$\left[\frac{3}{4}, \frac{3}{4}, x\right]$	[9,78]
10	$\left[\frac{1}{4}, -x, \frac{3}{4}\right]$	[10,56]
11	$\left[\frac{1}{4}, \frac{3}{4}, -x\right]$	[11,79]
12	$\left[\frac{3}{4}, x, \frac{3}{4}\right]$	[12,53]
13	$\left[\frac{1}{2} - x, \frac{3}{4}, \frac{1}{4}\right]$	[13,38]
14	$\left[\frac{1}{2} - x, \frac{1}{4}, \frac{3}{4}\right]$	[14,37]
15	$\left[\frac{1}{4}, \frac{1}{4}, x + \frac{1}{2}\right]$	[15,96]
16	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2} - x\right]$	[16,95]
17	$\left[\frac{3}{4}, \frac{1}{2} - x, \frac{1}{4}\right]$	[17,67]
18	$\left[\frac{1}{4}, x + \frac{1}{2}, \frac{1}{4}\right]$	[18,68]
19	$\left[\frac{1}{4}, \frac{1}{2} - x, \frac{3}{4}\right]$	[19,65]
20	$\left[\frac{3}{4}, x + \frac{1}{2}, \frac{3}{4}\right]$	[20,66]
21	$\left[x + \frac{1}{2}, \frac{1}{4}, \frac{1}{4}\right]$	[21,46]
22	$\left[x + \frac{1}{2}, \frac{3}{4}, \frac{3}{4}\right]$	[22,45]
23	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2} - x\right]$	[23,88]
24	$\left[\frac{3}{4}, \frac{3}{4}, x + \frac{1}{2}\right]$	[24,87]
25	$\left[\frac{1}{4}, x + \frac{1}{2}, \frac{3}{4}\right]$	[29,84]
26	$\left[\frac{1}{4}, \frac{3}{4}, x + \frac{1}{2}\right]$	[30,57]
27	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2} - x\right]$	[31,59]
28	$\left[\frac{3}{4}, \frac{1}{2} - x, \frac{3}{4}\right]$	[32,82]
29	$\left[\frac{3}{4}, \frac{1}{4}, x + \frac{1}{2}\right]$	[33,54]
30	$\left[\frac{1}{4}, \frac{1}{2} - x, \frac{1}{4}\right]$	[34,80]
31	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2} - x\right]$	[35,55]
32	$\left[\frac{3}{4}, x + \frac{1}{2}, \frac{1}{4}\right]$	[36,77]
33	$\left[\frac{1}{4}, \frac{3}{4}, x\right]$	[39,72]
34	$\left[\frac{3}{4}, \frac{3}{4}, -x\right]$	[40,71]
35	$\left[\frac{3}{4}, -x, \frac{3}{4}\right]$	[41,91]
36	$\left[\frac{1}{4}, x, \frac{3}{4}\right]$	[42,92]
37	$\left[\frac{1}{4}, -x, \frac{1}{4}\right]$	[43,89]
38	$\left[\frac{3}{4}, x, \frac{1}{4}\right]$	[44,90]
39	$\left[\frac{1}{4}, \frac{1}{4}, -x\right]$	[47,64]
40	$\left[\frac{3}{4}, \frac{1}{4}, x\right]$	[48,63]
41	$\left[x + \frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	[49,74]
42	$\left[x + \frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	[50,73]
43	$\left[\frac{1}{2} - x, \frac{1}{4}, \frac{1}{4}\right]$	[51,76]
44	$\left[\frac{1}{2} - x, \frac{3}{4}, \frac{3}{4}\right]$	[52,75]
45	$\left[-x, \frac{3}{4}, \frac{3}{4}\right]$	[61,86]
46	$\left[-x, \frac{1}{4}, \frac{1}{4}\right]$	[62,85]
47	$\left[x, \frac{1}{4}, \frac{3}{4}\right]$	[69,94]
48	$\left[x, \frac{3}{4}, \frac{1}{4}\right]$	[70,93]

Table 8: Wyckoff site: 96h, site symmetry: 1

No.	position	mapping
1	$[x, y, z]$	[1]

continued ...

Table 8

No.	position	mapping
2	$[x, -y, -z]$	[2]
3	$[-x, y, -z]$	[3]
4	$[-x, -y, z]$	[4]
5	$[z, x, y]$	[5]
6	$[y, z, x]$	[6]
7	$[-y, z, -x]$	[7]
8	$[-z, -x, y]$	[8]
9	$[-y, -z, x]$	[9]
10	$[z, -x, -y]$	[10]
11	$[y, -z, -x]$	[11]
12	$[-z, x, -y]$	[12]
13	$\left[\frac{1}{2} - x, z + \frac{1}{2}, \frac{1}{2} - y\right]$	[13]
14	$\left[\frac{1}{2} - x, \frac{1}{2} - z, y + \frac{1}{2}\right]$	[14]
15	$\left[\frac{1}{2} - z, \frac{1}{2} - y, x + \frac{1}{2}\right]$	[15]
16	$\left[z + \frac{1}{2}, \frac{1}{2} - y, \frac{1}{2} - x\right]$	[16]
17	$\left[y + \frac{1}{2}, \frac{1}{2} - x, \frac{1}{2} - z\right]$	[17]
18	$\left[\frac{1}{2} - y, x + \frac{1}{2}, \frac{1}{2} - z\right]$	[18]
19	$\left[\frac{1}{2} - y, \frac{1}{2} - x, z + \frac{1}{2}\right]$	[19]
20	$\left[y + \frac{1}{2}, x + \frac{1}{2}, z + \frac{1}{2}\right]$	[20]
21	$\left[x + \frac{1}{2}, \frac{1}{2} - z, \frac{1}{2} - y\right]$	[21]
22	$\left[x + \frac{1}{2}, z + \frac{1}{2}, y + \frac{1}{2}\right]$	[22]
23	$\left[\frac{1}{2} - z, y + \frac{1}{2}, \frac{1}{2} - x\right]$	[23]
24	$\left[z + \frac{1}{2}, y + \frac{1}{2}, x + \frac{1}{2}\right]$	[24]
25	$\left[x, y + \frac{1}{2}, z + \frac{1}{2}\right]$	[25]
26	$\left[x, \frac{1}{2} - y, \frac{1}{2} - z\right]$	[26]
27	$\left[-x, y + \frac{1}{2}, \frac{1}{2} - z\right]$	[27]
28	$\left[-x, \frac{1}{2} - y, z + \frac{1}{2}\right]$	[28]
29	$\left[z, x + \frac{1}{2}, y + \frac{1}{2}\right]$	[29]
30	$\left[y, z + \frac{1}{2}, x + \frac{1}{2}\right]$	[30]
31	$\left[-y, z + \frac{1}{2}, \frac{1}{2} - x\right]$	[31]
32	$\left[-z, \frac{1}{2} - x, y + \frac{1}{2}\right]$	[32]
33	$\left[-y, \frac{1}{2} - z, x + \frac{1}{2}\right]$	[33]
34	$\left[z, \frac{1}{2} - x, \frac{1}{2} - y\right]$	[34]
35	$\left[y, \frac{1}{2} - z, \frac{1}{2} - x\right]$	[35]
36	$\left[-z, x + \frac{1}{2}, \frac{1}{2} - y\right]$	[36]
37	$\left[\frac{1}{2} - x, z, -y\right]$	[37]
38	$\left[\frac{1}{2} - x, -z, y\right]$	[38]
39	$\left[\frac{1}{2} - z, -y, x\right]$	[39]
40	$\left[z + \frac{1}{2}, -y, -x\right]$	[40]
41	$\left[y + \frac{1}{2}, -x, -z\right]$	[41]
42	$\left[\frac{1}{2} - y, x, -z\right]$	[42]
43	$\left[\frac{1}{2} - y, -x, z\right]$	[43]
44	$\left[y + \frac{1}{2}, x, z\right]$	[44]
45	$\left[x + \frac{1}{2}, -z, -y\right]$	[45]
46	$\left[x + \frac{1}{2}, z, y\right]$	[46]
47	$\left[\frac{1}{2} - z, y, -x\right]$	[47]
48	$\left[z + \frac{1}{2}, y, x\right]$	[48]

continued ...

Table 8

No.	position	mapping
49	$\left[x+\frac{1}{2},\, y,\, z+\frac{1}{2}\right]$	[49]
50	$\left[x+\frac{1}{2},\, -y,\, \frac{1}{2}-z\right]$	[50]
51	$\left[\frac{1}{2}-x,\, y,\, \frac{1}{2}-z\right]$	[51]
52	$\left[\frac{1}{2}-x,\, -y,\, z+\frac{1}{2}\right]$	[52]
53	$\left[z+\frac{1}{2},\, x,\, y+\frac{1}{2}\right]$	[53]
54	$\left[y+\frac{1}{2},\, z,\, x+\frac{1}{2}\right]$	[54]
55	$\left[\frac{1}{2}-y,\, z,\, \frac{1}{2}-x\right]$	[55]
56	$\left[\frac{1}{2}-z,\, -x,\, y+\frac{1}{2}\right]$	[56]
57	$\left[\frac{1}{2}-y,\, -z,\, x+\frac{1}{2}\right]$	[57]
58	$\left[z+\frac{1}{2},\, -x,\, \frac{1}{2}-y\right]$	[58]
59	$\left[y+\frac{1}{2},\, -z,\, \frac{1}{2}-x\right]$	[59]
60	$\left[\frac{1}{2}-z,\, x,\, \frac{1}{2}-y\right]$	[60]
61	$\left[-x,\, z+\frac{1}{2},\, -y\right]$	[61]
62	$\left[-x,\, \frac{1}{2}-z,\, y\right]$	[62]
63	$\left[-z,\, \frac{1}{2}-y,\, x\right]$	[63]
64	$\left[z,\, \frac{1}{2}-y,\, -x\right]$	[64]
65	$\left[y,\, \frac{1}{2}-x,\, -z\right]$	[65]
66	$\left[-y,\, x+\frac{1}{2},\, -z\right]$	[66]
67	$\left[-y,\, \frac{1}{2}-x,\, z\right]$	[67]
68	$\left[y,\, x+\frac{1}{2},\, z\right]$	[68]
69	$\left[x,\, \frac{1}{2}-z,\, -y\right]$	[69]
70	$\left[x,\, z+\frac{1}{2},\, y\right]$	[70]
71	$\left[-z,\, y+\frac{1}{2},\, -x\right]$	[71]
72	$\left[z,\, y+\frac{1}{2},\, x\right]$	[72]
73	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z\right]$	[73]
74	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, -z\right]$	[74]
75	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, -z\right]$	[75]
76	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z\right]$	[76]
77	$\left[z+\frac{1}{2},\, x+\frac{1}{2},\, y\right]$	[77]
78	$\left[y+\frac{1}{2},\, z+\frac{1}{2},\, x\right]$	[78]
79	$\left[\frac{1}{2}-y,\, z+\frac{1}{2},\, -x\right]$	[79]
80	$\left[\frac{1}{2}-z,\, \frac{1}{2}-x,\, y\right]$	[80]
81	$\left[\frac{1}{2}-y,\, \frac{1}{2}-z,\, x\right]$	[81]
82	$\left[z+\frac{1}{2},\, \frac{1}{2}-x,\, -y\right]$	[82]
83	$\left[y+\frac{1}{2},\, \frac{1}{2}-z,\, -x\right]$	[83]
84	$\left[\frac{1}{2}-z,\, x+\frac{1}{2},\, -y\right]$	[84]
85	$\left[-x,\, z,\, \frac{1}{2}-y\right]$	[85]
86	$\left[-x,\, -z,\, y+\frac{1}{2}\right]$	[86]
87	$\left[-z,\, -y,\, x+\frac{1}{2}\right]$	[87]
88	$\left[z,\, -y,\, \frac{1}{2}-x\right]$	[88]
89	$\left[y,\, -x,\, \frac{1}{2}-z\right]$	[89]
90	$\left[-y,\, x,\, \frac{1}{2}-z\right]$	[90]
91	$\left[-y,\, -x,\, z+\frac{1}{2}\right]$	[91]
92	$\left[y,\, x,\, z+\frac{1}{2}\right]$	[92]
93	$\left[x,\, -z,\, \frac{1}{2}-y\right]$	[93]
94	$\left[x,\, z,\, y+\frac{1}{2}\right]$	[94]
95	$\left[-z,\, y,\, \frac{1}{2}-x\right]$	[95]

continued ...

Table 8

No.	position	mapping
96	$\left[z,\, y,\, x+\frac{1}{2}\right]$	`[96]`